Submit your source code file as well as a brief report with the following components:

- for each epsilon value in {0.01, 0.1, 1, 10}, report the MAE for your locally differentially private estimates

0.01 - 26505.324527005134

0.1 - 3054.8250709155704

1 - 242.39543407109826

10 - 11.998279241123116

- discuss the above observations and what you learn from both implementation exercises (HW 3 and 4)

Both HW3 and HW4 as the epsilon value increases MAE decreases. This is because as epsilon increases privacy decreases. In HW4 implementation epsilon impacts the odds of data points being swapped in the binary vectors. So an epsilon of 0 gives you a 50/50 chance of reporting the right/wrong value. If you gave an epsilon of 0 it would make the data completely random and have no use.